APPLICATION OF *F*-REGRESSION METHOD TO FUZZY CLASSIFICATION PROBLEM

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Abstract. In regression analysis, outliers always represent difficulties because they cause modeling errors. But under certain circumstances, they can actually contain useful information, as shown on the example of problem described in this article. That is why the task of outlier identification and analysis presents a twofold interest from the technical point of view. It is shown that *f*-regression method has good outlier detection capability and can be successfully applied to fuzzy classification problem.

Keywords: *f*-regression method, fuzzy regression, outliers, outlier detection, data clustering.

1 Introduction

Original idea of fuzzy regression analysis was proposed by Tanaka et al. in 1982 [6]. In his paper, Tanaka discussed fuzzy linear model for which fuzzy parameters were to be obtained using crisp inputs and either crisp or fuzzy outputs. This problem is effectively reduced to linear programming problem, which makes it easy to use and implement. A number of papers could be named in which application of conventional fuzzy regression analysis (also referred to as possibilistic regression analysis), as well as it improvements, is discussed. For instance, issue of outlier detection is discussed fairly often and one of the latest approaches which employs penalty coefficients can be found in [7].

In contrary, *f*-regression method [3] is a fuzzy regression method which is based on fuzzy input and output data and crisp parametric dependency between them. The key aspect of *f*-regression is its utilization of similarity measure between a model and a fuzzy point. Although much more complicated, this method has inherent capability to detect and analyze abnormal data, as shown below.

The problem of *f*-regression analysis is formulated as follows. Suppose to be given crisp parametric dependency

$$f(\boldsymbol{x},\boldsymbol{a}) = 0, \tag{1}$$

where

$$\boldsymbol{x} = (\boldsymbol{t}, y)^T, \boldsymbol{t} = (t_1,\ldots,t_{n-1})^T \tag{2}$$

($\boldsymbol{t}$ stands for vector of independent variables, y is dependent variable) and N fuzzy input/output data points. We aim at finding such parameter vector $\boldsymbol{a}$ for which f would fit experimental points best, in a certain sense.

Fuzzy input/output data are obtained from measurements. Each value is extended from crisp number to fuzzy number using subject area knowledge. The fuzzy numbers being used are symmetric L-type fuzzy numbers with membership function μ_A

$$\mu_A(t) = L\left(\left|\frac{t-a}{\alpha}\right|\right). \tag{3}$$

Here, fuzzy set A is described by two values: a is the *mean value* and $\alpha > 0$ is the *spread*. (For $\alpha = 0$ we get the formal representation of crisp numbers). L-function used in practical implementation has the following representation

$$L(u) = \exp\left(-u^m\right), m \geq 1 \tag{4}$$

Parameter m is called *form parameter* and its value defines the overall form of all fuzzy numbers: for $m \approx 1$, the number looks more like triangular fuzzy number; $m \approx 10$ gives fuzzy numbers corresponding to uniformly distributed random values. For $m \approx 2$, we obtain fuzzy numbers resembling normally distributed random numbers.

Values for m and α are chosen based on subject area knowledge and/or common sense considerations. Generally, the more uncertainty is present, the greater the α parameter has to be.

f-regression is capable of handling of two types of fuzzy points (by fuzzy point we mean a vector of fuzzy numbers):

- symmetric L-type fuzzy points, which were initially described in [1]. But here, production T-norm is applied to fuzzy coordinates X_i instead of min T-norm to obtain membership function for fuzzy points:

$$\mu_Q(\boldsymbol{x}) = \prod_i \mu_{X_i}(x_i) \tag{5}$$

- in certain circumstances, there is a necessity to take interactions between different variables into account. For this case, elliptic fuzzy points were proposed based on work of Celmi š [3]:

$$\mu_Q(\boldsymbol{x}) = L\left(\left[(\boldsymbol{x}-\bar{\boldsymbol{x}})^T A^{-1}(\boldsymbol{x}-\bar{\boldsymbol{x}})\right]^{\frac{1}{2}}\right), \tag{6}$$

where $\bar{\boldsymbol{x}}$ denotes the vector of individual coordinates' mean values, also called point's *apex*, and A is $n\times n$ positive definite matrix referred to as *panderance matrix*. As its name implies, α-cut of elliptic fuzzy point is ellipsoid. Matrix A is calculated based on spread values and *interrelation coefficients* associated with desirable interrelation effect between variables.

The key *f*-regression concept is similarity measure. It is used to determine the degree to which fuzzy point belongs to a crisp curve. A similarity measure of fuzzy point with respect to parametric curve is equal to maximum of fuzzy point's membership function on the set of points belonging to the curve.

$$M(\boldsymbol{a}) = \sup_{f(\boldsymbol{x},\boldsymbol{a})=0} \mu_Q(\boldsymbol{x}) \tag{7}$$

The tractability of a similarity measure is straightforward: if it is (near) 1, we say that point (almost) lies on a curve; in the case of 0, the point does not belong to that curve. With respect to regression analysis, in the case of *zero* similarity we say that a fuzzy point does not fit explicit parametric dependency for that given parameter values.

For linear regression curve, analytical solution to (7) has been obtained for both types of fuzzy points and has the following form

$$M(\boldsymbol{a}) = L(d(\boldsymbol{a})) \tag{8}$$

with

$$d(\boldsymbol{a}) = \frac{\left|\bar{y} - \left(a_0 + \boldsymbol{a}^T \bar{\boldsymbol{t}}\right)\right|}{\left(\alpha_y^{\frac{m}{m-1}} + \sum_j^{n-1} \left|\alpha_{t_j} a_j\right|^{\frac{m}{m-1}}\right)^{\frac{m-1}{m}}} \tag{9}$$

for symmetric *L*-type fuzzy points and

$$d(\boldsymbol{a}) = \frac{\left|\bar{y} - \left(a_0 + \boldsymbol{a}^T \bar{\boldsymbol{t}}\right)\right|}{\left(\boldsymbol{a}^T A \boldsymbol{a}\right)^{1/2}} \tag{10}$$

for elliptic points.

For those curves which are not linear but linear in the parameters, a transformation can be made which allows using formulas (9) and (10). This will be discussed in detail in section 4.

For non-linear and implicit curves (i.e. for most general cases of (1)) no analytical solution exists, though an approach using numeric methods has been proposed to accomplish that. Unfortunately, this topic falls out of the scope of current paper.

The second key concept of *f*-regression is aggregation operator. We introduce family of *weighted mean-based aggregation operators* that collect information about model curve passing through all fuzzy experimental data. Three members of this family are of interest to us due to their practical applicability: *geometric mean*, *arithmetic mean* and *quadratic mean*.

$$MG(\boldsymbol{a}) = \prod_i M_i^{w_i}(\boldsymbol{a}) \text{ s.t. } \sum_i^N w_i = 1 \tag{11}$$

- *Geometric mean* is a strong generalization operator. It means that when applied to similarity measures, this operator tends to give a degree of conformance of all fuzzy points with the model at the same time. In other words, if this operator is maximized over parameter space, we get a model which fits **all** fuzzy points at the same time – like the least squares method;

$$MA(\boldsymbol{a}) = \sum_i w_i M_i(\boldsymbol{a}) \text{ s.t. } \sum_i^N w_i = 1 \tag{12}$$

- *Arithmetic mean* is a compensatory operator. That is, applying this operator to similarity measures, one gets an idea about what is the average degree of similarity measure of all fuzzy points with the model. When maximized over parameter space, this operator yields a model that passes through the **majority** of points;

$$MQ(\boldsymbol{a}) = \left(\sum_i^N w_i M_i^2(\boldsymbol{a})\right)^{\frac{1}{2}} \text{ s.t. } \sum_i^N w_i = 1 \tag{13}$$

- *Quadratic mean* is an 'elitist' operator. This is expressed in a fact that individual good-fitting points add more to the value of this operator than does 'gray mass' of average-fitting points (opposite to geometric mean logic). When maximized, this operator tends to find model that passes through 'elite' **group** of points.

As one can see from (11)-(13), the aggregation operator is bound to [0; 1] and thus can be interpreted as *degree of membership* of solution to the fuzzy set of best solutions for the problem.

So the solution of fuzzy regression analysis problem is effectively reduced to unconstrained optimization problem, optimal solution for which will be parameter vector

$$a^* = \arg\max_{a \in \mathrm{R}^n} MP(a) \qquad (14)$$

2 Outlier detection

A note should be made that the similarity measure M effectively serves as a criteria for outlier detection. An outlier is defined as follows:

Definition 1. [2] An outlier is the one that appears to deviate markedly from other members of the collected dataset.

That is why, if the majority of the points shows high level of similarity measure M_i and some point has value of M_i very close to 0, it is right to say that this point conforms to given **Definition 1**. It *deviates markedly* from other members in the sense that it does not belong to the function which fits all other points.

We introduce threshold value h, which is used in taking decisions about whether point should be regarded as outlier or not. This value should lie within boundaries (0; 0.5). Value 0.5 is referenced because a situation when similarity measure of point to a curve is 0.5 is regarded as undetermined – the point belongs to the curve at the same degree as it does not belong to it. In practical cases, threshold value $h = 0.05$ can be safely taken.

It is necessary to say that different aggregation operators have different attitudes toward outliers:

- *geometric mean* has behavior similar to that of LSM, it perceives fuzzy data as a whole. As a side effect, it assigns lower similarity measures for points that deviate most significantly from others. Moreover, it can be shown that if we assume crisp inputs, fuzzy outputs with standard deviation value for spread α and form parameter $m = 2$, analysis will produce the same result as for conventional LSM;
- *arithmetic mean* is the one most often used in practical implementations since its robustness to outliers. Theoretically it can provide solutions in cases when up to 50% of data are outliers;
- *quadratic mean* has a very special behavior and mainly can be used to effectively spot small group of points. It is well described by the fact that for 10 experimental points, it is more favorable for MQ to pass only through 3 points with degree 1.0 (others with 0.0) than to pass through all 10 with degree 0.5.

3 Technical problem

Consider the following problem (it is taken from "Robust regression and outlier detection" by Rousseeuw and Leroy, 1987, J. Wiley). We analyze the dependency between body mass and brain mass of mammals. In our study we use dataset, which consists of 65 different mammals, ranging from small (mole, rat) to big (cow, elephant). In the dataset we also put some pre-historical animals – dinosaurs, – and primates.

We seek to fit the dependency with function:

$$b = a_0 \cdot B^{a_1}, \qquad (15)$$

where b and B are brain and body masses, respectively.

Let's put in some common sense considerations. We know that a human has larger brain then, say, a dog of the same weight. More generally, 'body – brain' curve for primates will lie higher then average mammal curve. We don't change the form of analytical dependency; we just choose different parameters for primates. The reverse consideration is true for dinosaurs. It is important to note that one can solve this problem qualitatively: mind intuitively captures data samples that significantly deviate from the general mass. The computer cannot because it has no internal knowledge on how to group and classify data samples. So we must teach it.

We use our knowledge about subject area: since the data are average values for each species, we must fuzzify them. A good approach would be to set a range on body weight and brain weight values to form a rectangle of all possible cases. But we should also reflect the fact that smaller individuals have smaller brain and vice versa by setting *interrelation coefficient* between these variables. Therefore we should represent each experimental data as an elliptic fuzzy point, with α spread value equal to 0.5 of the absolute value for both b and B and high interrelation coefficient.

Considering the nature of experimental data, which is average brain and body mass, we should set form parameter $m = 2$ to reflect its probabilistic origin.

The described considerations were used to input data to `FuReA` software tools. The workflow was designed with a goal to minimize human effort. The analytical form of equation (15) was not changed in the course of analysis.

First, we used *geometric mean* on data. The real world question asked was: can we obtain the most general dependency between mammal's body and brain masses? The results so obtained were unsatisfactory; the value of aggregation operator *MG* was nearly zero. This could potentially mean two things: *a)* the spreads chosen were too small; *b)* data contains samples wildly different from the majority. We dropped *a)* because we made competent decision on spread configurations based on subject area knowledge.

Second, we used *arithmetic mean*. The real world question asked this time was: what is the body – brain dependency for majority of mammals and what mammals don't fit in? This operator has provided us with a clearer picture: about 20% of points (in quantity) were discarded as outliers (threshold was 0.05). Thus we got a confirmation on previous *b)* hypothesis. Due to compensatory property of arithmetic mean we obtained a model fitting as much points as it can: Rodents, Felidae, Canis, Artiodactyla – all fell into this category. The rest of points could be either *a)* inconsistent, inaccurate data or *b)* data too hard to fit in general category or even *c)* data forming special categories. Since we know the data is accurate we proceed without *a)*.

Last, we switched to *quadratic mean*. The question now was formulated in the following way: among animals not falling into general category, is there a group described by specific dependency? To proceed with this question, we excluded all 'general category' data from our dataset (effectively setting its weight to 0). The analysis yielded a specific dependency for a small category of primates (chimpanzee, baboon, human etc.), and a number of points being outliers with respect to this category. The *c)* hypothesis and a common sense fact described in the beginning of this paragraph were proven: primates' body – brain curve lies higher then that of general category mammals. If we exclude primates from dataset we can once again try to find a different category. The category thus obtained is dinosaurs (tiny brain size).

The result is shown on Figure 1. For convenience, the chart placed here is in logarithmic coordinates.

For reference, the results obtained from Least Squares Method (LSM) are represented by thin black line. One can observe that it does not seem to fit data very well. The results of *MA* run of *f*-regression are depicted in dark line, running through rhomb points. One can see that the majority of mammals actually fit this dependency, forming the largest category, except for some 20% data samples.

Primates are denoted by gray squares and their specific dependency is represented by gray line which is clearly a special dependency within their category. It is also interesting to note, however, that certain mammals, like galago which is biologically close to primates, reveals high membership to this category.

Gray triangles denote the other miscellaneous data which did not fall into two main categories. They are dinosaurs, opossum and tenrec. They were categorized together by *MQ* aggregation operator.

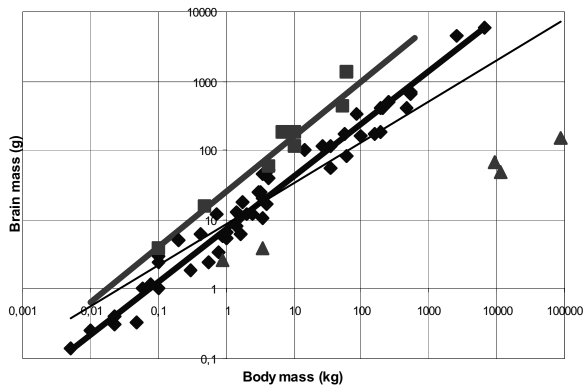

Figure 1: Result of regression analysis

4 Discussion

Software tools

f-regression method is implemented in software tools for fuzzy regression analysis called `FuReA` [4]. All the experiments described in this paper were carried out using these tools.

Nonlinear regression approach

We address nonlinear curve (15) described in the paper by transforming it to model linear in the parameters

$$\ln m = \ln a_0 + a_1 \ln M \; . \qquad (16)$$

One should note that since *m* and *M* are fuzzy variables, ln *m* and ln *M* are also fuzzy and should be calculated using fuzzy transformation approach (this transformation is done internally by the software and does not require and special action from the user).

There are a lot of different model classes that are or can be made linear in their parameters with a certain transformation. For example, all polynomial models fall into this category. However, solving a system where not real variables but their transformations are taken, we intentionally distort the original problem and produce "problem falsification" [3].

To obtain linear model that we know how to handle we introduce vector of variables

$\boldsymbol{v} = (v_1,...,v_c)$ and transformation functions $\boldsymbol{h}(\boldsymbol{x}) = (h_1(\boldsymbol{x}),...,h_c(\boldsymbol{x}))$:

$$\boldsymbol{v} = \boldsymbol{h}(\boldsymbol{x}) \tag{17}$$

In new variable space we again obtain function linear w.r.t. $\boldsymbol{v}$ and $\boldsymbol{a}$ by applying Taylor series expansion to $\boldsymbol{h}(\boldsymbol{x})$ and dropping terms of second and higher order, namely

$$\boldsymbol{v} = \boldsymbol{h}(\bar{\boldsymbol{x}}) + J(\boldsymbol{x} - \bar{\boldsymbol{x}}) + o(\boldsymbol{x} - \bar{\boldsymbol{x}}) \tag{18}$$

Here J is $c\times n$ Jacobian matrix of functions $\boldsymbol{h}(\boldsymbol{x})$:

$$J : j_{lk} = \frac{\partial h_l(\boldsymbol{x})}{\partial x_k}, l = 1,...,c, k = 1,...,n \tag{19}$$

Now, let's discuss the case when we have model linear w.r.t. $\boldsymbol{v}$. To get experimental data for this transformed problem we must apply (18) to every experimental point. From fuzzy set theory we know that the result of operations on fuzzy numbers have to be calculated through application of Zadeh's extension principle. The only two unambiguous operations are multiplication by a constant and addition of two fuzzy numbers:

$$\begin{aligned} &\gamma X(a,\alpha) = X(\gamma a,|\gamma|\alpha),\\ &X(a,\alpha) + Y(b,\beta) = Z(a+b,\alpha+\beta). \end{aligned} \tag{20}$$

Substituting expressions (20) into (18), we calculate transformed experimental data.

A note on the precision of formula (18) should be made. The quality of approximation can be poor if the magnitude of the second order derivatives is relatively large compared to first order derivatives and spreads of variables. In some cases this approximation error can be undesirably large due to high curvature of transformation functions near the expansion point.

Search method used in `FuReA`

Procedure of searching for optimal solution is different for different types of aggregation operator. The following steps are common for all *MP*s:

1. Conventional least squares method is used to solve the problem to obtain value $\boldsymbol{a}_{LSM}$.
2. $\boldsymbol{a}_{LSM}$ is used as a starting point for a local optimization method, for example, Simplex or Nelder-Mead technique.

These two steps are sufficient to obtain $\boldsymbol{a}^*$ which is optimal solution for *MG* aggregation operator because it has only one global extremum. Arithmetic mean and quadratic mean operators require additional steps due to their highly non-linear attitude and multiple extrema.

3. Additional search is taken to determine a number of $\boldsymbol{a}$ with high values of aggregation operator.
4. Each of vectors obtained on the previous step is used as a starting point for local optimization.

For step 3, *combinatorial search* method developed especially for that purpose is use.

The main idea behind combinatorial search method is that we can obtain a fixed (though sometimes very large) number of solutions (essentially parameter vectors) for the problem, at least one of which is guaranteed to be so close to the global optimum that global optimum will be obtained from it after local optimization.

Suppose to be given N crisp points on $\boldsymbol{x} \in R^n$. Then for each index set $\boldsymbol{c} = (c_1...,c_n)$, $1 \le c_i \le N, c_i \ne c_j \; \forall i \ne j$, i,j=1,…,n, we can find unique parameter vector $\boldsymbol{a}(\boldsymbol{c}) : M_{Q_{c_i}}(\boldsymbol{a}(\boldsymbol{c})) = 1 \forall c_i$ from system of linear equations (if it is non-singular, i.e. if $\boldsymbol{x}_{c_i} \ne \boldsymbol{x}_{c_j} \; \forall i \ne j$), the value of *MA* aggregation operator being $MA(\boldsymbol{a}(\boldsymbol{c})) \ge \frac{n}{N}$.

Evidently $MA(\boldsymbol{a}(\boldsymbol{c})) > \frac{n}{N}$ means that other $\boldsymbol{a}(\boldsymbol{c})$ lie near that point. We introduce a procedure to find $\tilde{\boldsymbol{c}} : MA(\boldsymbol{a}(\tilde{\boldsymbol{c}})) = \max_{\boldsymbol{c}} MA(\boldsymbol{a}(\boldsymbol{c}))$. Moreover, we are seeking for a solution which provides a better solution then least squares method does, namely $MA(\boldsymbol{a}(\tilde{\boldsymbol{c}})) \ge MA(\boldsymbol{a}_{LSM})$. That way we can be sure that we apply effective filter to get a starting point for local optimization method which is guaranteed to lead to *the same or better* solution then $\boldsymbol{a}_{LSM}$ leads to.

One immediate drawback, however, is that one gets a total of C_n^N combinations between which choice is to be made. That is why some heuristics must be introduced to decrease this number, for example, by dropping obviously ineligible combinations. This can be achieved using the fact that *MA* values for different index sets $\boldsymbol{c}$ are not completely independent.

5 Conclusion

In conclusion we want to summarize the main results of this investigation:

– *f*-regression can approach data analysis problem from qualitative point of view as well as from traditional quantitative approach;

- *f*-regression method produces desirable results when adequately calibrated and the right aggregation operator is chosen;
- *f*-regression outlier detection capability has advanced to a level of fuzzy classification;
- *f*-regression provides means to take additional information about subject area into account;
- the results obtained by *f*-regression method correspond to common sense as it was illustrated on a simple one input – one output problem. This induces the like capabilities for much more complex problems which cannot be illustrated on a graph.

6 References

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